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# Evaluation of physicochemical, tribological and oxidative stability properties of chemically modified rice bran and karanja oils as viable lubricant base stocks for industrial applications

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ABSTRACT

The lubricating properties of the two vegetable oils with distinct free fatty acid content were studied in the work i.e., Rice bran oil (RBO) with low free fatty acids and Karanja oil (KO) with high free fatty acids. The main aim of the present work is to improve the oxidative stability of RBO and KO, using the transesterification and epoxidation method. The physicochemical, tribological, and oxidative stability of pure and modified oils were evaluated as per standards. The study revealed the effectiveness of epoxidation over the transesterification process on the selected oils.

## 1. Introduction

Lubricants help in reducing friction and wear of surfaces that come into touch with one another. They are also utilized to dissipate heat and foreign particles formed at the work zone. Most of the lubricants used in the industries are mineral oil-based, whereas vegetable oil-based lubricants constitute only 0.1% of the market [1]. The mineral oil-based lubricants are non-renewable, a concern for fossil fuel depletion also causes pollution and health hazards [2], hence bio-based lubricants are getting more important nowadays. Many scientists have examined various vegetable oils (VO) as a potential base oil for lubricants to address these difficulties.

Bio-lubricants can be made from both edible and non-edible oils. Vegetable oils possess superior lubricating properties and stable thermal properties. They exhibit 70–100% biodegradability [3], low toxicity, high flash point, and high viscosity [4]. The primary reason for the less utilization of vegetable oil-based lubricants in the industrial sector is the lack of its oxidative stability and poor low-temperature properties [5]. Several studies were conducted on the various vegetable oils as the base oil for industrial lubricants like rapeseed oil [6], palm kernel oil [7,8], castor oil [8], coconut oil [8], mustard oil [9], sunflower oil [6,10], Pongamia [11], cottonseed oil [12], canola oil [13], sesame oil [14], jatropha oil [15], Tilapia oil [16] and Calophyllum inophyllum [17]. A few researchers have already used chemical modifications to enhance the shelf life of oils. Epoxidation, transesterification, and hydrogenation are some of the leading chemical pathways used to enhance the oxidation stability of oils. All of these reactions aim to lower the number of unstable sites in vegetable oils' triglyceride structure.

In the present study, the transesterification and epoxidation processes are performed to alter the oil structure chemically. The rearrangement of an ester's alkyl group with another alkyl group from alcohol is performed during the transesterification procedure. In the first stage of transesterification, the triglycerides react with 3 moles of methanol to produce methyl esters, and glycerol is formed as a by-product. The bio lubricant is then made by reacting the methyl esters with trimethylolpropane (TMP). TMP is a polyol containing three hydroxy functional groups. It is found that the Polyols possess higher thermal stability [18].

During the epoxidation procedure, the triglycerides with percarboxylic acids are used to break the carbon-carbon double bonds resulting in the formation of a new epoxide functional group. There have been a few publications on the transesterification of oils and some of them are reviewed below. Palm oil methyl ester with TMP had been synthesized, and the thermal properties such as flash and fire points were found to be improved [19]. Palm oil-based TMP ester has shown improved tribological properties [20]. Jatropha oil methyl ester was reacted with TMP at different temperatures, and the optimum temperature was found to be 150 ℃ with a ratio of 3.9:1 for the reactants [21]. Jatropha curcas

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oil-based bio lubricant using TMP was synthesized. The pour point of the product was observed as low as −30 °C; the flashpoint was > 300 °C and viscosity was 79 cP [22]. The TMP esters of rapeseed oil are used as a base oil for bio-hydraulic fluid and results were compared with conventional hydraulic fluids. The results have shown good cold stability, tribological properties, and resistance against oxidation at elevated temperatures [23]. TMP was used to transesterify palm oil-based methyl esters to polyol esters in a study. The presence of methyl esters was found to improve wear properties and resulted in superior oxidative stability [24]. A study on epoxidation of vegetable oils showed that the coefficient of friction (COF) of epoxidized sunflower and soybean oils was reduced than that of mineral and synthetic oils [25]. Comparing pure soybean oil, epoxidized soybean oil showed better thermal properties and wear properties. The cause for friction and wear decrease has been related to the cross-linking of epoxide rings on the surface of metals, which reduces friction between moving parts [26]. The epoxidized Rice bran oil has shown a considerable improvement in tribological and lubricating performance than pure oil. The tribological characteristics and oxidation of epoxidized RBO were found to be superior. The elimination of unsaturation from the triglyceride structure increased the oxidative stability of epoxidized oils substantially [27]. A similar trend was observable even on rapeseed oil [28].

The work presented here aims to analyze the lubricating characteristics of derived TMP-based tri-ester and epoxidized bio-lubricants from Karanja oil (KO) and Rice bran oil (RBO), such as physicochemical, tribological, and oxidation stability. Karanja is a non-edible oil, and their trees are fast-growing even in the drought-prone area. While India is the world's largest producer of rice, and rice bran is commonly available for oil extraction, it also includes gamma-oryzanol, a natural antioxidant. The lubricant properties of the vegetable oils depend on the fatty acid composition and free fatty acid content. The effect of chemical modification processes on lubricating properties of vegetable oils with varied free fatty acid content is not yet reported. The RBO and KO chosen for the present study have a varying level of free fatty acids and the effect of chemical modification on these oils is attempted. The lubricant properties of modified KO and RBO samples were evaluated and compared to pure oil samples. The results showed that the modified oils outperformed pure oils in terms of lubricating characteristics. According to the study, the modified vegetable oils were also discovered to show potential as a stoke for bio-lubricants and to be a viable energy-saving base-stock for industrial lubricants.

## 2. Materials and methods

The methodology followed for chemical modifications and lubricant property evaluation of RBO and KO is explained clearly in the following section. All of the experiments were repeated three times and the average readings were recorded.

### 2.1. Fatty acid composition

The vegetable oil's fatty acid composition contributes significantly to good lubricating qualities, better low-temperature behavior, and improved oxidative stability. A gas chromatography-mass spectroscopy approach was utilized to determine the types of fatty acids present in vegetable oil. The test was performed on a DB wax column. The test samples were transformed into fatty acid methyl esters and put into the apparatus, which was kept at 250°C. The percentage composition of fatty acids present in the oil samples is determined using the mass spectroscopy instrument.

### 2.2. Synthesis of Rice bran and Karanja oil-based methyl esters

In this study, the transesterification procedure for RBO and KO was performed as followed, 200 ml of RBO/KO were taken into a conical flask and heated to 100 ℃ in a hot air oven for 20 min; this is done to get rid of the moisture in the oil. 3 g of KOH was added to 55 ml of methanol, and the KOH-Methanol mixture is swirled manually until the KOH dissolves in the methanol. The heated oil along with KOH-Methanol mixture was then transferred to a 500 ml container fitted with a condenser, thermometer, and stirrer. Stirring is performed on the sample and a temperature of 65 °C is applied for 1 h. To remove the glycerol component a separating funnel was used and the reaction products get separated after 6 h. To remove contaminants, the product was washed three times in warm water (70 °C). Rice Bran oil methyl ester (RBOME) and Karanja oil methyl esters (KOME) were obtained from the water-washed mixture stored in a separating funnel The obtained oil is vacuum heated to remove any remaining water content. The synthesis of methyl esters is depicted in Fig. 1.

### 2.3. Synthesis of Rice bran and Karanja oil-based TMP esters

The reaction of TMP with RBOME and KOME is explained below. 100 ml of RBOME/KOME taken into a flask fitted with a vacuum pump, thermometer, and stirrer. The reaction of synthesizing TMP esters from methyl esters is shown below in Fig. 2.

The methyl ester was exposed to a temperature of 100 °C, and TMP was used in the amount of 33 g. 1 g of sodium methoxide was added as a catalyst after the TMP had melted then the sample was blended for 2 h using a magnetic stirrer. The methanol was removed from the sample by suction created by the vacuum pump. The sample was then water washed with warm water (70 °C). The TMP esters of RBO and KO synthesized at the end of the transesterification process are here onwards called TRBO and TKO. Fig. 3 depicts the outline of the transesterification technique used in this study.

### 2.4. Synthesis of epoxidized Rice bran and Karanja oil

The basic concept of epoxidation reaction for both RBO and KO is shown below in Fig. 4.

In this study, the epoxidation is done by taking 100 ml of vegetable oil samples. 5 ml acetic acid and 3–4 drops of $H_2SO_4$ were mixed into the oil sample. Then the mixture is swirled manually and the required quantity of $H_2O_2$ was mixed drop-by-drop in a closed atmosphere. The epoxidation reaction of the oil sample is carried out under vacuum conditions with constant magnetic stirring and heating at 70 ± 5 °C for 4.5 h. The reaction mixture is then transferred into a separating funnel. A clear liquid separation will be formed at the bottom of the funnel. This bottom liquid layer needs to be removed. The oil formed at the top of the separating funnel is then washed five times using hot deionized water. The temperature of deionized water was kept at around 70 °C. The unreacted reagents present in the oil sample were removed along with the deionized water using the same separating funnel. The final obtained epoxidized oil sample is then heated using an electric heater with a vacuum pump set up to remove the excess water from the final oil samples. The epoxidized samples of RBO and KO synthesized at the end of the epoxidation process are here onwards called ERBO and EKO. Fig. 5 depicts the epoxidation technique used in this study.

### 2.5. Fourier transform infrared (FTIR) spectroscopy

Fourier transform infrared (FTIR) spectroscopy was used to evaluate the functional groups of epoxidation and transesterified oil samples using Thermoscientific Nicolet iS50 with software for data recording. The spectrum was obtained in the range of 400–4000 $cm^{-1}$. For all the tests conducted, a Nicolet iS50 FTIR Spectrometer with a KBr beam splitter, PbSe detector and a Nicolet iS50 FTIR spectrophotometer module was employed. The excitation source was a laser with a current of 0.66 A and an IR source was used with 9 A. OMNIC Software was used to examine the spectra.

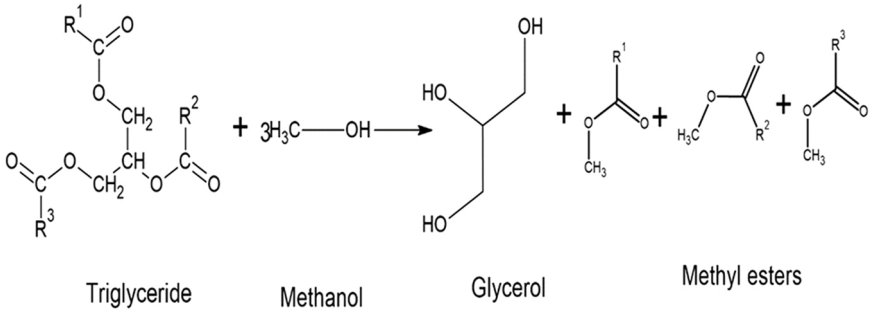



**Fig. 1.** Synthesis of methyl esters by transesterification.

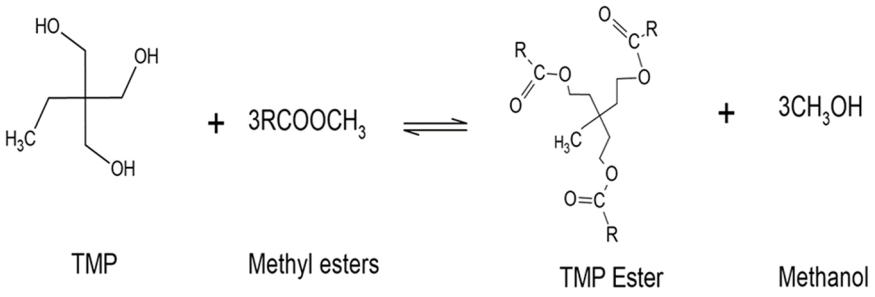



**Fig. 2.** Conversion of methyl esters to TMP esters.

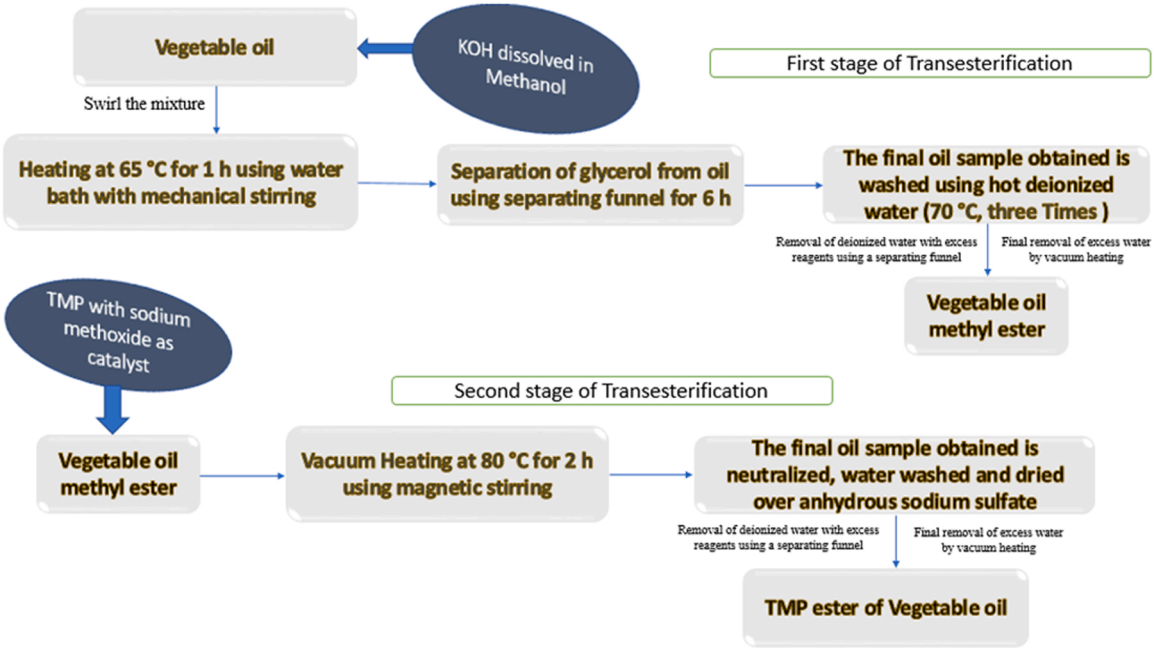



**Fig. 3.** Outline of the transesterification reaction.

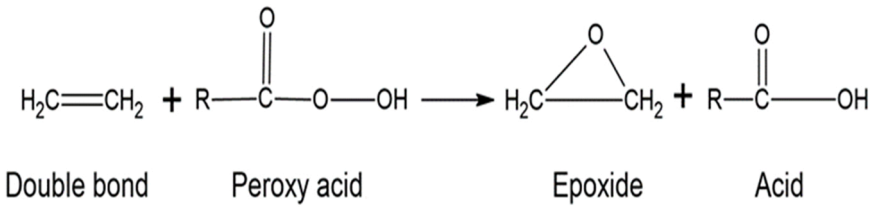



**Fig. 4.** Epoxidation reaction.

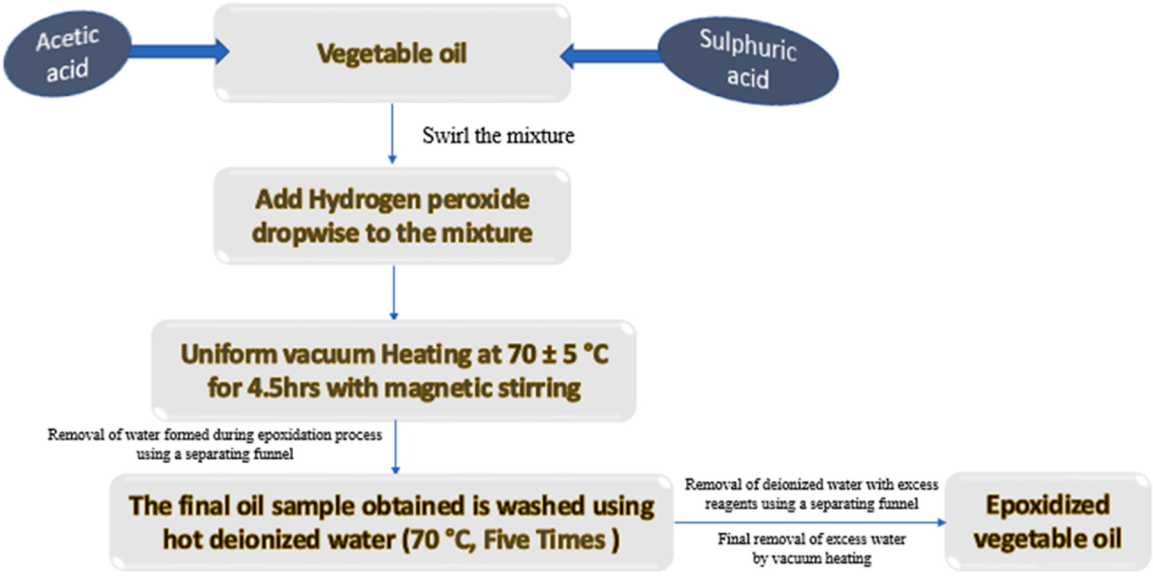



**Fig. 5.** Outline of the Epoxidation Reaction.

### 2.6. Tribological properties

A Four-Ball Tester is used to evaluate the tribological properties such as the coefficient of friction (COF) and wear scar diameter (WSD) of the samples. The testing was performed as per the ASTM D4172 standard. The four ball tester consists of a vertical spindle with a ball collet which is rotated by an AC motor. The three balls are fitted in the ball pot with a ball locking ring and lock nut using a torque wrench. The pot is filled with sample oil. The ball pot is placed on top of the antifriction disc and connected to a heater. The normal load of 392 N was applied to the balls by a loading lever. The balls are made up of chrome steel with a 12.7 mm diameter and hardness of 64HRC. The test was conducted with a rotating speed of 1200 rpm at 75 °C for a period of 1 h. The wear scar diameter of the fixed balls was measured using an optical microscope and the worn surface images were obtained using Leica optical microscope (10x, 100 μm).

### 2.7. Physico-chemical properties

The kinematic viscosity of the sample is measured using a Cannon-Fenske Opaque Viscometer in accordance with ASTM D446. In a Cannon-Fenske tube, a sample of oil is obtained and maintained in the viscometer bath. After the oil has reached the requisite temperature, the time it took for it to achieve the marking must be recorded, and the viscosity must be assessed. Metal-to-metal contact will increase as viscosity decreases [29].

Titration methods based on Indian standard IS: 548 (Part 1) –1964 were used to analyze chemical properties such as acid, peroxide, and iodine value. The acid number and peroxide value indicate the product's shelf life and oxidation resistance. The acid number is used to evaluate the amount of free fatty acids present in the oil. Whereas the peroxide value gives the amount of peroxides formed during the oxidation process. Both the acid and peroxide values need to be at lower concentrations for better shelf life. The level of unsaturation in the sample is indicated by the iodine number. Lower iodine values are desirable to avoid oxidation caused by the unsaturated fatty acids.

### 2.8. Oxidative stability

Vegetable oils have low oxidative stability due to fatty acids with many double bonds, which is one of the major disadvantages. All of the oil samples were subjected to a Hot Oil Oxidation Test (HOOT) in accordance with the AOCS Cd-12–57 standard. The samples were kept in a hot air oven for 120 h at 100 degrees Celsius. The viscosity variation after 120 h was evaluated at 40 °C using a kinematic viscometer. The oil sample's oxidative stability will be indicated if the viscosity variation is low.

The oxidation stability of the oil samples is also evaluated by using a rotating pressure vessel oxidation stability test (RPVOT) apparatus as per ASTM D2272. The oil sample is taken into a glass container and then filled with reagent water along with a freshly cleaned catalyst coil. The glass container is covered with a PTFE disk and inserted into a vessel body. The vessel body is tightened with the vessel cap and oxygen is supplied inside with a pressure of 6.2 bar. The heating bath filled with silicone oil is maintained at 150 °C. The vessel with the test sample is submerged in the heating bath and continuous uniform rotation is maintained. The test is complete after the pressure drops more than 1.75 bar below the maximum pressure and the time taken for the pressure drop is noted as induction time.

### 2.9. Thermogravimetric analysis

The thermogravimetric analysis (TGA) determines how a sample's weight changes as a function of temperature. This approach is used to determine the decomposition and oxidation of a material. The test was conducted in an oxygen environment. The percentage of weight loss and gain were measured at required temperatures. The occurrence of oxidation is indicated by an increase in weight gain, whereas weight loss explains thermal stability in terms of mass degradation.

### 2.10. Energy consumption

During the machining operation, energy consumption has to be reduced for efficient production. All of the oil samples were tested for energy consumption using a four-ball test. The energy conservation equation was used to calculate the thermal energy and Eq. (1) is shown below [30,31].

$$\text{Thermal energy(TE)} = \frac{fWgr}{1000} \tag{1}$$

Where, $f$ = Coefficient of Friction, $W$ = applied load (kg), $g$ = gravitational acceleration (m/s$^2$), and $r$ = distance between the axis of rotation to distance from the center of the contact surface on the lower balls in mm (i.e., 3.67 mm).

### 2.11. Flash temperature parameter

The Flash temperature parameter determines the temperature at which the oil vaporizes or ignites and fails (FTP). The FTP was introduced by Lane to determine the possibility of lubrication failure during operation[30–32]. The FTP was evaluated for pure and chemically modified oils. The equation for FTP was formulated for four-ball tester and shown below in Eq. (2).

$$\text{Flash Temperature Parameter(FTP)} = \frac{w}{d^{1.4}} \quad (2)$$

Where $w$ = applied load (kg) and $d$ = mean WSD (mm).

### 2.12. Cold flow properties

The low-temperature properties such as pour and cloud points were assessed using ASTM D97 and ASTM D2500, respectively for all the samples. These values are especially important during the winter months and while using lubricants in cold climates. Hence the cloud and pour points of a suitable lubricant should be low enough.

## 3. Results and discussions

The percentage composition of saturated and unsaturated fatty acids of RBO and KO are shown in Table 1. However, monounsaturated fatty acids are found to be the major constituents in both oils. Saturated fatty acids give extremely high oxidation stability as they don't possess double bonds, but at room temperature, they are in a solid-state and provide poor low-temperature behavior [33,34]. Monounsaturated fatty acids offer the optimum balance of oxidative and thermal stability, also exceptional tribological and low-temperature qualities [35]. For a superior bio-lubricant formulation, a blend of saturated and unsaturated fatty acids is suggested. From Table 1, it was observed that oleic acid is the most abundant component in both RBO and KO, accounting for 41.50% and 51.55%, respectively. RBO and KO had similar percentages of saturated and unsaturated fatty acids, however, RBO had somewhat more poly unsaturation than KO.

The presence of the unsaturated fatty acids and the highly unstable β hydroxyl group in the glycerol molecule are the main reasons for the low oxidation stability of oil [18]. The β-CH present in the vegetable oils is highly reactive and possesses low thermal and oxidative stability. Efforts are made to remove the β carbon, unstable hydroxyl groups and convert double bonds into stable compounds from the triglycerides by chemical modification.

The FTIR was conducted on the chemically modified oils to confirm the transesterification and epoxidation reactions. The functional groups after the reaction were identified using FTIR spectroscopy and the FTIR spectrum of epoxidized and transesterified RBO are explained in Fig. 6. Similarly, the FTIR spectrum of epoxidized and transesterified KO are explained in Fig. 7. The major peaks identified for TRBO and TKO are at 1435 $cm^{-1}$, and 1436 $cm^{-1}$ respectively, which was due to the presence

**Table 1**
Fatty acid composition of RBO and KO.

| Name of acid | Nature | RBO Content (%) | KO Content (%) |
|---|---|---|---|
| Oleic Acid (C18:1) | Monounsaturated | 41.50 | 51.55 |
| Linoleic Acid (C18:2) | Poly unsaturated | 32.74 | 16.39 |
| Linolenic Acid (C18:3) | Poly unsaturated | 0.07 | 3.54 |
| Palmitic Acid (C16:0) | Saturated | 20.12 | 10.66 |
| Stearic Acid (C18:0) | Saturated | 2.16 | 6.78 |
| Eicosanoic Acid (C20:1) | Saturated | 0.97 | 1.85 |
| Lignoceric acid(C24:0) | Saturated | – | 2.24 |
| Behenic acid (C22:0) | Saturated | – | 5.81 |
| Total Saturated acid | | 23.25 | 27.34 |
| Total Unsaturated acid | | 74.31 | 71.48 |
| Total Polyunsaturated acid | | 32.81 | 19.93 |

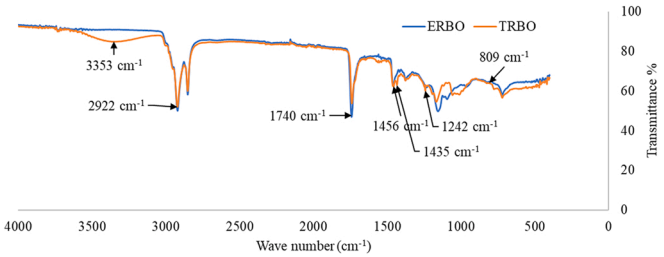



**Fig. 6.** FTIR spectra of ERBO and TRBO.

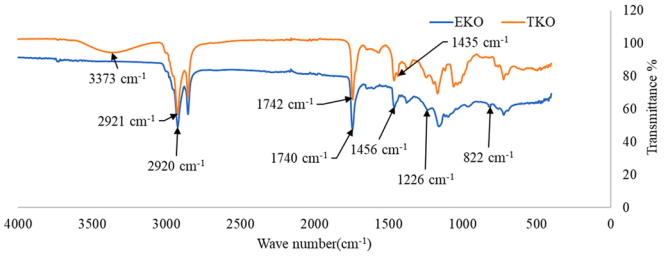



**Fig. 7.** FTIR spectra of EKO and TKO.

of the $CH_3$-O group after the transesterification reaction.

The major peaks identified for ERBO and EKO at 1226–1233 $cm^{-1}$, and 822 $cm^{-1}$, respectively are because of the presence of epoxy and the epoxide group after the epoxidation reaction. Other notable peaks can be found at 2920–2921 $cm^{-1}$, 1740–1742 $cm^{-1}$, and 1456–1461 $cm^{-1}$, which correspond to the alkene, ester, and alkane, respectively. The FTIR results confirmed the presence of the $CH_3$-O group and epoxides after the chemical modification of the oil samples.

Fig. 8 illustrates the chemical properties of samples under the study. From Fig. 8, it is understood that the free fatty acids, which are measured by acid value were tremendously reduced after the transesterification for both TKO and TRBO, which is a sign of good oxidation stability [36]. Lower acid values are desirable for longer oil shelf life. However, both ERBO and EKO showed just a little drop in acid levels. From Fig. 8, it is also evident that the peroxide values have drastically reduced after the transesterification and epoxidation. Following the chemical modification, the triglyceride structure's β -carbon and double bonds were removed, which inhibited auto-oxidation and reduced the principal oxidation products known as peroxides.

Lower peroxide values are preferable for improved base oil oxidation stability [37]. It is understood that pure KO contains higher free fatty acids and the acid value was reduced significantly after transesterification. Whereas pure RBO contains lower free fatty acids hence both transesterification and epoxidation were found to be effective in reducing the acid value. However, peroxide values were reduced for RBO and KO after both transesterification and epoxidation.

Fig. 9 shows the iodine value, which is a measure of unsaturation in oils. The iodine value was found to be reduced for both the oils after the transesterification and epoxidation.

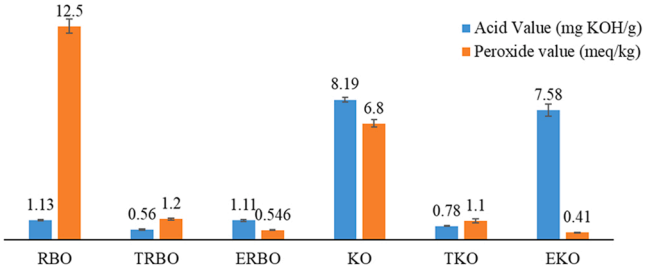



**Fig. 8.** Acid and peroxide values of pure and chemically modified oils.

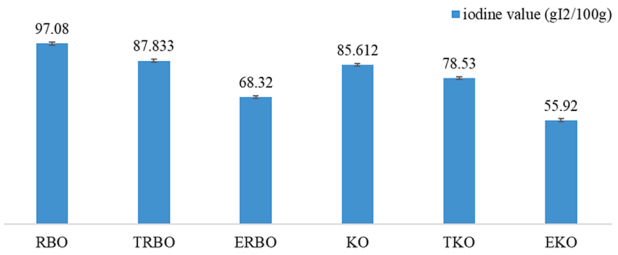



**Fig. 9.** Iodine values of pure and chemically modified oils.

However, the reduction in iodine value was more after the epoxidation. When compared to pure oils, TRBO and TKO had an iodine value reduction of 9.52% and 8.27%, respectively. When compared to pure oils, ERBO and EKO showed a reduction of 29.62% and 34.69%, respectively. Hence epoxidation has resulted in lesser unsaturation in the base oil this is because of the conversion of double bonds into oxirane rings. The chemical properties evaluated revealed that transesterification will effectively inhibit auto-oxidation by reducing free fatty acid content [38]. Whereas, epoxidation will improve the oxidation stability by removing unsaturation in the oil sample.

To accomplish its purpose, a good lubricant should have a lower coefficient of friction and a smaller wear scar diameter. Figs. 10 and 11 depict the tribological properties of pure, transesterified, and epoxidized oils, such as COF and WSD.

From Figs. 10 and 11, it is clear that ERBO and EKO have shown the lowest COF and WSD this is because of the presence of oxirane rings which may crosslink and adhere to the metal surface [28]. However, the COF and WSD of transesterified samples were found to be higher than pure and epoxidized samples. After the transesterification reaction, the glycerol part will be eliminated from the oil which dissociates the carboxylic acid (COOH), and resulted in higher COF and WSD when compared with pure and epoxidized oils. The COOH is a polar head present in the pure RBO and KO that adheres to the metal surface and resulted in a lower COF and WSD.

The pure vegetable oils contain a polar carboxyl group (COOH) at the end of fatty acid which adheres to the metal surface and provides better tribological properties to the oil which is schematically shown in Fig. 12.

Whereas in the case of epoxidized vegetable oils the presence of COOH and oxirane rings which are polar and has cross-linking capability can be seen. Hence the tribological properties are better than pure oils because of the synergetic effect of carboxylic acid and oxirane ring adhering. The schematic representation of tribo film in the epoxidized oils is shown in Fig. 13.

In the case of transesterified oils, the glycerol part is replaced with a polyol ester and the polarity difference of them has resulted in poor tribological properties. The wear scar images of pure and modified rice bran and karanja oils are shown in Fig. 14.

From Fig. 14 it is clear that transesterified oils have shown more wear grooves when compared to pure and epoxidized oils which are in accordance with COF and WSD results. In the case of epoxidized oils, the

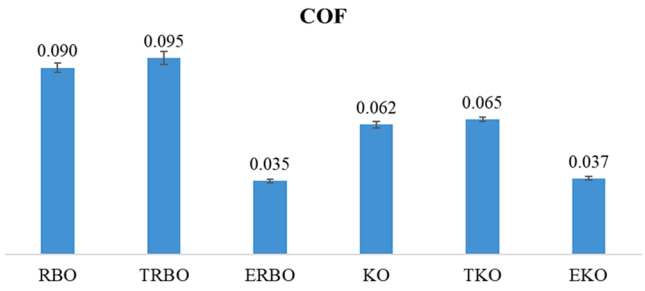



**Fig. 10.** Coefficient of Friction of pure and chemically modified oils.

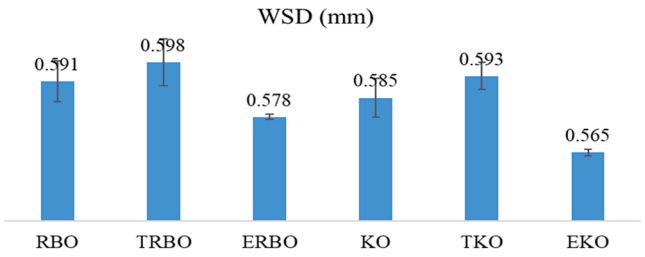



**Fig. 11.** Wear Scar Diameter of pure and chemically modified oils.

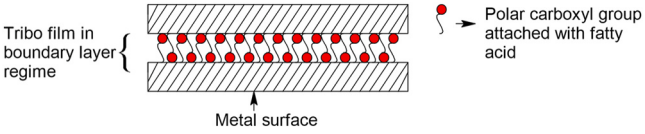



**Fig. 12.** Schematic representation of tribo film in pure oils.

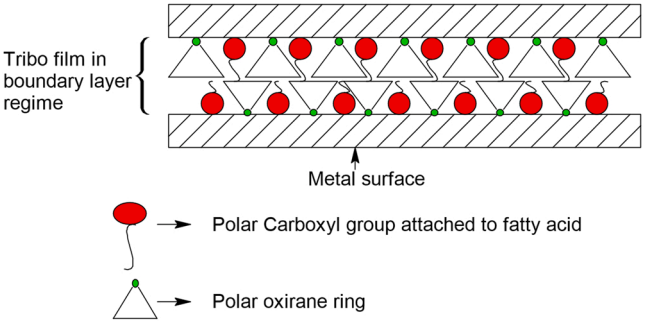



**Fig. 13.** Schematic representation of tribo film in epoxidized oils.

presence of the COOH group and oxirane rings have resulted in fewer grooves than other samples, and the mechanism is already explained in Fig. 13. The abrasive wear mechanism was observed for all the samples, whereas for pure KO both adhesive and abrasive wear mechanism was identified.

The lubricant's ability is determined by its viscosity, which is a measure of the oil's resistance to flow. The kinematic viscosity of pure and modified oils was evaluated and shown in Fig. 15. The degree of unsaturation has a vital role in the rheological behavior of oils, monounsaturated fatty acids (MFA) results in higher viscosity, whereas polyunsaturated fatty acids (PFA) results in lower viscosity of the lubricants [35].

This explains the viscosity trend in Fig. 15, where KO which contains more MFA and lesser amounts of PFA has exhibited slightly higher viscosity. Because the glycerol part from the oil was removed, the viscosity was found to be slightly lowered after transesterification. The epoxidized oils have shown very high viscosity values when compared with the pure and transesterified oils this is because of the formation of epoxy and epoxide groups. Epoxidized oil samples can be used at high-temperature applications [39].

The HOOT test was used to evaluate the oxidation stability of the samples in terms of viscosity variation and the findings are reported in Table 2. From Table 2, it is understood that the viscosity increments are higher for pure oils than for modified oils indicating better oxidation stability of modified oils.

This could be due to lower levels of free fatty acids and unsaturation, which prevent the creation of higher molecular weight primary and secondary oxidation products. However, epoxidized samples of both the oils have shown the least viscosity increment of all the samples under study indicating higher oxidation stability. Compared to other samples, ERBO has shown the least viscosity increment of 16.09%, implying the

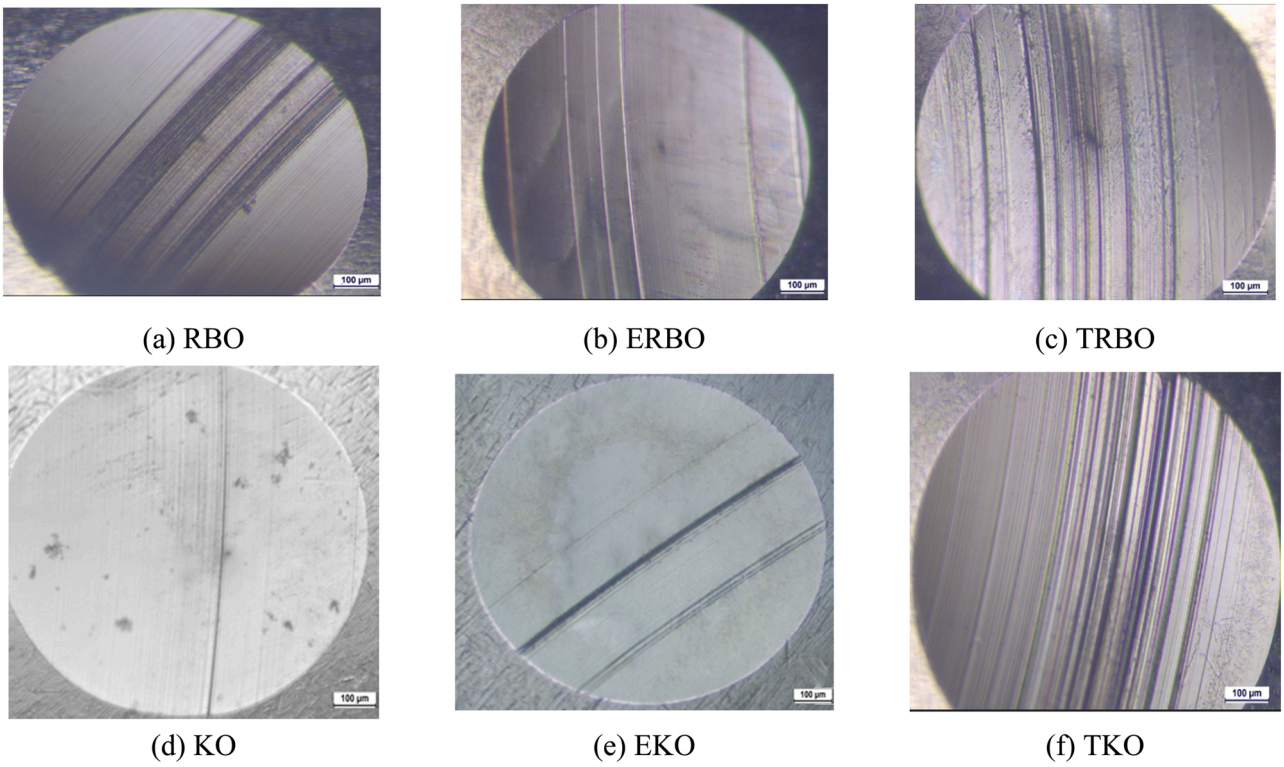



**Fig. 14.** Worn surface image of pure and chemically modified oils after tribological test.

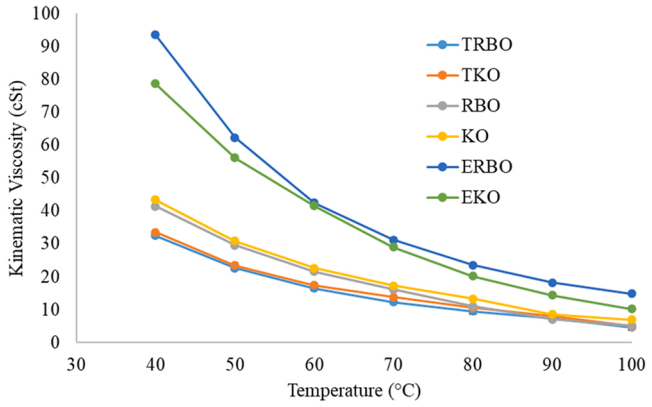



**Fig. 15.** Kinematic viscosities of pure and chemically modified oils at various temperatures.

**Table 2**
Kinematic viscosity variation for pure and chemically modified oils after HOOT.

| Samples | Viscosity increment (%) | RPVOT induction time (min) |
|---|---|---|
| RBO | 19.30 ± 0.28 | 10.11 ± 0.15 |
| KO | 17.45 ± 0.24 | 13.40 ± 0.11 |
| TRBO | 17.32 ± 0.34 | 14.21 ± 0.09 |
| TKO | 16.72 ± 0.37 | 19.32 ± 0.13 |
| ERBO | 16.09 ± 0.13 | 22.47 ± 0.17 |
| EKO | 16.35 ± 0.26 | 21.17 ± 0.21 |

highest oxidation stability. This is because of the removal of unsaturated bonds in the oil sample and also RBO possesses a natural anti-oxidant called γ-oryzanol [40]. The oxidation stability is also evaluated for pure and modified oils in terms of the induction time by using the RPVOT apparatus and the results are shown in Table 2. The RPVOT results are in accordance with the results obtained from the HOOT test. The viscosity increment after the HOOT and induction time of mineral oil-based lubricant was observed to be 15.61 ± 0.11% and 30.13 ± 0.17 min which is slightly better than modified vegetable oils.

The TGA was conducted on the pure, modified oils and the results are depicted in Figs. 16 and 14. It can be understood from Fig. 16 that the mass degradation of ERBO started at 380 °C. Whereas TRBO and RBO started at 180 °C and 250 °C respectively.

RBO and ERBO have a higher mass degradation than TRBO at higher temperatures. Fig. 17 shows that mass deterioration of EKO and KO began at 350 degrees Celsius. TKO, on the other hand, began at 220 °C. As a result, it is obvious that epoxidized oils have superior thermal stability to pure and transesterified oils.

Energy consumption of pure and chemically modified oils was evaluated and shown in Fig. 18. ERBO and EKO have shown the least energy consumption when compared with all the samples. Since energy consumption has a relation with COF and both ERBO and EKO have shown the least COF due to the tribofilm formed by oxirane rings.

For pure and modified oils, the flash temperature parameter (FTP) was calculated, and the results are given in Fig. 19. The higher the FTP number, the greater the lubricating capabilities and the less likely the

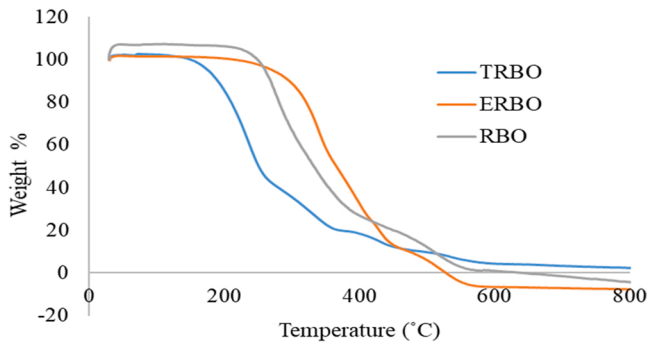



**Fig. 16.** TGA of pure and chemically modified RBO.

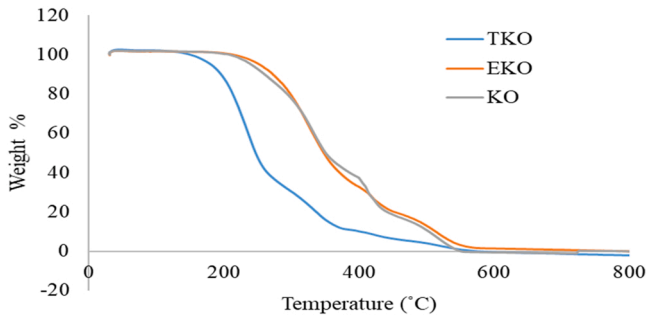



**Fig. 17.** TGA of pure and chemically modified KO.

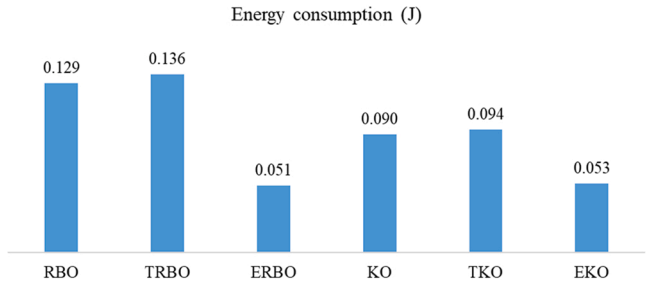



**Fig. 18.** Energy consumption of pure and chemically modified oils.

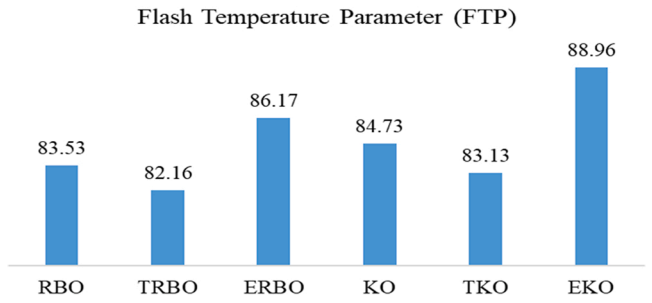



**Fig. 19.** Flash Temperature Parameter of pure and chemically modified oils.

lubrication layer will break down. The FTP was found to be higher for EKO followed by ERBO. Since FTP has an inverse relation with WSD, EKO which has lower WSD compared with all the samples under study has shown higher FTP.

From the present study, it is understood that the tribological, rheological properties and oxidation stability of the pure oils were drastically improved after the chemical modification process. The modified vegetable oils can be a viable source of bio-lubricant, according to the FTP and energy consumption.

The cloud and pour points for all the samples were evaluated and the results are shown in Table 3. From the Table 3 it is clear that the pure vegetable oils had shown better cold flow properties than chemically modified oils, this is because of the excess macrocrystalline formations in modified oils at low temperatures which restricts the flowability of oils [41]. Also, the increase in saturation levels in modified oils leads to straight fatty acids which do not contain bent configuration causing close packing of molecules and resulting in poor cold flow properties [33].

## 4. Conclusions

The shelf life or oxidative stability of the vegetable oils is the major drawback that needs to be addressed through chemical modification. The transesterification and epoxidation of RBO and KO were investigated in this work, and the following conclusions were reached:

**Table 3**
Cold flow properties of pure and modified oil.

| Sample name | Cloud point | Pour point |
|---|---|---|
| RBO | -5 | -8 |
| TRBO | -3 | -6 |
| ERBO | -1 | -3 |
| KO | 7 | 3 |
| TKO | 6 | 3 |
| EKO | 8 | 5 |

- The transesterification and epoxidation reactions are confirmed by FTIR spectroscopy, and peaks of the $CH_3$-O group are found in TRBO and TKO. Also, the epoxy and epoxide peaks are present in ERBO and EKO.
- Modified oils are shown to have better acid, peroxide, and iodine values than pure oils. The acid values were found to be reduced after the transesterification procedure, whereas the peroxide and iodine content was reduced after the epoxidation.
- The COF of epoxidized oils has shown the least values of pure and transesterified oils this is because of the tribofilm formed by cross-linking of oxirane ring with the metal surface.
- The transesterified oils have shown slightly lower viscosity values than pure oils. Whereas, epoxidized oils have shown higher viscosities than pure oils.
- The epoxidized oils have shown higher oxidation and thermal stability when compared to pure and modified vegetable oils. The RPVOT and the HOOT results indicated that the oxidation stability of modified vegetable oils needs to be further enhanced to compete with commercial oils.
- The epoxidized oils have also shown the least energy consumption and higher FTP values compared to all the oil samples under the study.
- The cold flow properties were found to be better for pure oils than chemically modified oils because of the excess crystal formation in modified oils.
- The major drawback of vegetable oils has been addressed by chemical modification methods. The comparative study revealed that epoxidation is a better method to chemically modify vegetable oils. Transesterification, on the other hand, is better suited to oils with a high free fatty acid content.

## CRediT authorship contribution statement

**Sneha Edla, Ananthan D. Thampi**: Conceptualization. **Ananthan D. Thampi, Pranav Prasannakumar, Rani S.**: Methodology. **Sneha Edla, Ananthan D. Thampi, Pranav Prasannakumar**: Investigation. **Sneha Edla, Ananthan D. Thampi**: Writing – original draft preparation. **Sneha Edla, Rani S**: Writing – review & editing. **Rani S.**: Supervision.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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